

Day 1: Question bank

1. Which of the following best defines a computer?

- A. A machine that stores only numbers
- B. A device that processes data into information
- C. A machine that cannot be programmed
- D. A device used only for games

Answer:

2. What is a program?

- A. A hardware device
- B. A set of instructions for the computer
- C. An operating system
- D. A collection of wires and circuits

Answer:

3. Which is not a reason for using programming languages?

- A. To instruct the computer to perform tasks
- B. To communicate with other humans
- C. To automate processes
- D. To create applications

Answer:

4. Which of the following is an example of a high-level language?

- A. Assembly
- B. C
- C. Machine code
- D. Binary

Answer:

5. What is “syntax” in programming?

- A. The logical steps of a program
- B. The rules for writing code in a language
- C. The storage capacity of a program
- D. The speed of code execution

Answer:

6. An algorithm can be best described as:

- A. Computer hardware
- B. A step-by-step procedure to solve a problem
- C. A computer virus
- D. A type of syntax error

Answer:

7. Which is the lowest level of programming language?

- A. Assembly language
- B. High-level language

- C. Machine language
- D. Script language

Answer:

8. Assembly language is:

- A. Directly understood by computers
- B. Written using mnemonics
- C. A type of high-level language
- D. Faster than machine code

Answer:

9. Which translator converts assembly language into machine language?

- A. Compiler
- B. Assembler
- C. Interpreter
- D. Linker

Answer:

10. Which translator converts high-level language into machine code all at once?

- A. Compiler
- B. Interpreter
- C. Assembler
- D. Loader

Answer:

11. Which translator converts and executes code line-by-line?

- A. Compiler
- B. Interpreter
- C. Assembler
- D. Linker

Answer:

12. Which is not an advantage of high-level languages?

- A. Easier to learn and write
- B. Hardware-independent
- C. Requires less time to debug
- D. Executes faster than machine language

Answer:

13. What does “machine language” consist of?

- A. Mnemonics
- B. Natural English words
- C. Binary codes (0s and 1s)
- D. Flowcharts

Answer:

14. Which of these is not a programming language?

- A. Python
- B. HTML
- C. Java
- D. C

Answer:

15. Why do we need translators for programming languages?

- A. To reduce the number of programmers
- B. To make code readable by humans
- C. To convert human-readable code into machine-readable form
- D. To store the code in the cloud

Answer:

16. Which of the following is true for compilers but not interpreters?

- A. They execute code line-by-line
- B. They generate a separate executable file
- C. They are slower for large programs
- D. They do not detect syntax errors

Answer:

17. Which one is an example of an interpreter-based language?

- A. Java (JVM execution)
- B. Python
- C. C
- D. Assembly

Answer:

18. In programming, an "error due to breaking syntax rules" is called:

- A. Logic error
- B. Runtime error
- C. Syntax error
- D. Semantic error

Answer:

19. Which format specifier is used to read an integer value using scanf()?

- A. %c
- B. %f
- C. %d
- D. %lf

20. Which of the following is NOT a valid identifier?

- A. _count
- B. sum1
- C. roll_no
- D. total marks

Lab Assignment: Practical C Programming

Instructions for Submission:

Code: Write the complete C source code for each question below.

Output: Run your code, take a clear screenshot of the output terminal and paste it directly underneath your code.

Format: Save this document as YourName_Lab1.pdf

1. Create a program to declare two integer variables, assign them values, and display their values.
2. Create a program to input name of the person and respond with *"Welcome NAME to C Programming"*.
3. Define variables for storing a user's first name, last name, and age using appropriate naming conventions and then display them.
4. Create a program to print the area of a square by inputting its side length.
5. Create a program to define a constant for the mathematical value pi (3.14159) and use it to calculate and print the circumference of a circle with a radius input from user.
6. Create a program to print the area of a circle by inputting its radius.
7. Create a program to swap two numbers.
8. Write a C program that asks the user to input two integers, then performs and displays the results of the following arithmetic operations:

Addition

Subtraction

Multiplication

Division (show the result as a float)

Modulus (remainder)

Make sure the program properly handles integer inputs and prints the results clearly.

Sample Output:

Enter two integers: 10 3

Results:

Addition = 13

Subtraction = 7

Multiplication = 30

Division = 3.33

Modulus = 1

9. Write a program to calculate Simple Interest using the formula
 $\text{interest} = (\text{principal} * \text{rate} * \text{time}) / 100$
10. Write a program to add, subtract, multiply, and divide two numbers.
11. Write a program to calculate Compound Interest.
 $\text{amount} = p * \text{pow}((1 + r / 100), t)$
 $\text{ci} = \text{amount} - p$; Hint: Use `#include <math.h>`
12. Write a program to calculate BMI (Body Mass Index).

bmi = weight / (height * height)

13. Write a program to convert Celsius to Fahrenheit. **f = (c * 9 / 5) + 32**

14. Write a program to convert Fahrenheit to Celsius. **c = (f - 32) * 5 / 9**

15. Write a program to calculate percentage of marks (5 subjects, out of 100 each).

total = m1 + m2 + m3 + m4 + m5

percentage = total / 5

16. Write a program to calculate the discounted price of an item.

discount = (price * discountPercent) / 100

finalPrice = price - discount

17. Write a program to calculate speed, given distance and time.

speed = distance / time

18. Write a program to calculate **net salary = (basic pay + allowance – deductions).**

19. **Predict the output and verify it by running the program.**

```
#include <stdio.h>
int main()
{
    int x = 5;
    printf("x = %d\n", x);
    printf("x++ = %d\n", x++);
    printf("After x++ = %d\n", x);
    printf("++x = %d\n", ++x);
    printf("Final x = %d\n", x);
    return 0;
}
```

20. **Predict the output and verify it by running the program.**

```
#include <stdio.h>
int main()
{
    int i = 6, j = 3, k;
    k = ++i * j--;
    printf("i = %d\n", i);
    printf("j = %d\n", j);
    printf("k = %d\n", k);
    return 0;
}
```

Explain:

1. Why is C called a mid-level programming language?
2. What are the features of the C programming language?
3. Explain internal execution of program to swap two numbers?
4. When your program is ready to execute?
5. Explain Compilation process in C.
6. Difference between compiler and interpreter
7. What is the purpose of #include <stdio.h> in a C program?
8. Explain the difference between printf() and scanf() with examples.
9. What are format specifiers in C? Explain any five with examples.
10. Explain the difference between pre-increment (++i) and post-increment (i++) with an example.
11. Why do we use the main() function in every C program?
12. What is the purpose of the return 0; statement?
13. Why do we use the address operator (&) in scanf()?
14. What is the difference between float and double?
15. Explain the difference between unary operators and arithmetic operators with examples.
16. What is a variable in C? Explain the rules for naming (identifiers) a variable with examples.
17. What are literals in C? Differentiate between constants and literals with examples.
18. What are keywords in C? Why can't keywords be used as variable names? Explain with examples.
19. What is the purpose of the void keyword in C? Explain its different uses with examples.
20. Differentiate between variables, constants, and literals with suitable examples.